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CPT: Differentiable Constraint Projection for Amortized, Zero-Shot Retargetable Steering of Autoregressive Generative Models

The per-step surrogate targets a concentrated distribution the sequence-level optimum does not

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ABSTRACT

Steering an autoregressive generative model to satisfy a quantitative constraint is usually posed at the level of whole sequences, where the Kullback–Leibler-optimal solution is an exponential tilt of the base distribution. We instead solve a constrained projection of the next-token distribution at every step, differentiably, and train through it. Two results follow, both measured on an exactly enumerable synthetic task and on crystal structure generation over MP-20. First, the per-step surrogate is not a worse approximation of the tilt but an approximation of a different distribution, and the difference favours it: at equal satisfaction of the expectation constraint, the surrogate's charge variance is 1.547 against the exact optimum's 10.452, and the optimum in fact *raises* variance 22.1% above the unconstrained data it starts from, because buying a mean of zero from an off-target distribution is cheapest by moving mass to both extremes. Individual samples from the surrogate therefore violate the constraint 1.42 times less often than samples from the optimum it approximates. Second, one trained checkpoint retargets across a whole family of constraint values with no retraining, statistically indistinguishable from seven separately trained models (two one-sided tests, $p = 0.0023$ at a pre-registered margin of 0.189) at one seventh of the training compute. The layer costs 6.0× the training wall clock of masking alone and admits an analytic Jacobian. We report where the method stops: steering saturates at a ceiling set by the backbone, two correlated constraints sharing one token budget compose only partially and asymmetrically, the constraint class is bounded by what a serialization makes determinable from a prefix, and best-of-n sampling beats the projection on accuracy given enough inference compute while collapsing output diversity to do so.

KEYWORDS controlled generation; constrained decoding; differentiable optimization layers; expectation constraints; crystal structure generation

1 Introduction

A practitioner who wants a generative model to respect a quantitative property, such as a target density, a charge balance, or a budget, is asking for something the model was not trained to provide. The standard formulation answers at the level of whole sequences: among all distributions whose expectation of some statistic ϕ meets a target, take the one closest in Kullback–Leibler divergence to the model. That distribution is an exponential tilt of the base and the target of distributional policy gradient (Sec. 2). It is also awkward to reach, because the tilt is defined over complete sequences while an autoregressive model emits one token at a time.

This paper takes the other route. At every decoding step we solve a small constrained projection of the next-token distribution: the closest distribution, in the same divergence, whose expected increment keeps a running budget on track. We train through that solve. The projection is a convex problem with an analytic Jacobian, so it is a layer rather than a decoding heuristic, in the sense established by the differentiable-optimization literature surveyed in Sec. 2.

The obvious objection is that a per-step surrogate must be worse than the sequence-level optimum, and on divergence from that optimum it is: we measure a fixed overhead of about one nat. The central result of this paper is that this framing is wrong about what the overhead buys. Measured against a model actually trained toward the tilt, the surrogate does not approximate the same distribution less well; it approximates a *different* distribution, one in which the constrained quantity is concentrated rather than spread. The sequence-level optimum, reaching a mean of zero from an off-target base, does so by moving mass to both extremes, so it raises the variance of the very quantity being constrained, above the unconstrained model's, and its individual samples violate the constraint more often than the surrogate's do (Fig. 1). Whether that trade is wanted depends on whether "most samples satisfy the constraint" or "the average satisfies it" is the requirement, and for most practitioners asking for a constraint it is the former.

The second result is about cost rather than accuracy. Because the constraint enters as a target value rather than as a training condition, one trained checkpoint can be deployed at constraint values it never saw, and it performs no worse than a family of separately trained models.

Contributions.

1. A differentiable per-step constraint projection with an analytic Jacobian, trained through rather than applied at decoding time (Sec. 3).
2. The concentration result: the surrogate targets a distribution whose constrained quantity is concentrated, while the sequence-level optimum anti-concentrates it (Sec. 4).
3. Zero-shot retargeting: one checkpoint covers a constraint family with statistically bounded equivalence to per-target training (Sec. 5).
4. Four scope boundaries, each measured rather than asserted: a steering ceiling, partial and asymmetric two-constraint composition, a serialization determinability limit, and inference-time baselines that beat the projection on accuracy at a diversity cost (Secs. 6 and 7).

> KEY TAKEAWAY
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> The per-step surrogate is not a cheaper approximation of the sequence-level optimum. It

targets a different distribution, one whose constrained quantity is concentrated rather than spread, and on the tasks measured here that distribution is closer to what a request for a constraint usually means.

2 Related work

Sequence-level distributional control. Posing controlled generation as a moment constraint and solving for the KL-minimal distribution gives an exponential family optimum, trained toward by distributional policy gradient [1]. That distribution is the reference object of this paper: our comparator arm is trained toward it, and our central claim is a statement about how it differs from what a per-step projection targets.

Per-step constrained decoding. Lexically constrained decoding enforces predicate-logic constraints during search without retraining [2], with lookahead heuristics added to reduce myopia [3]. GeLaTo instead distills a tractable generative model to supply exact conditional guidance [4], and sequential Monte Carlo methods reweight partial sequences against a target potential [5, 6]. These share the per-step locus of this work but treat the constraint as a filter or a reweighting; we solve a convex program whose solution is a distribution, and differentiate it. Ahmed et al. [7] are the nearest training-time neighbour, relaxing a logical constraint into a pseudo-semantic loss over a local approximation, the same shape of move as reducing a sequence-level expectation to a per-step one, though approximate in the constraint rather than exact in the support.

Differentiable optimization layers. Embedding an argmin in a network and backpropagating through its solution is established for quadratic programs [8], general convex programs [9], and hard constraint completion with correction [10]. Our layer is a specialization: a KL projection onto an affine slice of the simplex intersected with a support mask, small enough that the Jacobian is analytic rather than obtained by implicit differentiation of a generic solver.

Expectation versus per-instance constraints. The distinction our concentration result turns on is the one constrained Markov decision processes make between constraining an expectation and constraining every trajectory [11], refined by state-wise formulations [12] and by chance-constrained programming [13]. The precedent closest in mechanism is the safety layer, which projects a proposed action onto a feasible set at every step [14]. We adopt that vocabulary: the sequence-level tilt is an expectation constraint, and what our projection delivers is closer to a per-step feasibility guarantee whose aggregate happens to meet the expectation.

Crystal structure generation. Generative models for inorganic crystals include diffusion over invariant representations [15, 16], autoregressive language models over text serializations [17, 18], and large-scale diffusion trained for property targeting [19]. We use the same MP-20 subset of the Materials Project [20] and a text serialization in the spirit of the autoregressive line, because a serialization is what makes a per-step budget definable at all. Sec. 6 shows this is a real constraint on the method rather than an implementation detail.

3 Method

Let $p_\theta(\cdot \mid x_{<t})$ be the model's next-token distribution over a vocabulary $\mathcal{V}$, let $m_t \in \{0,1\}^{|\mathcal{V}|}$ be a mask marking tokens the grammar permits after the prefix $x_{<t}$, and let $\phi \in \mathbb{R}^{|\mathcal{V}|}$ assign each token an increment of the constrained quantity. Given a target total c for a sequence of length L, define the running budget

$$b_t = \frac{c - \sum_{s<t} \phi_{x_s}}{L - t + 1}, \tag{1}$$

the per-step increment that would land the total on target if every remaining step matched it. The layer solves

$$q_t = \arg\min_{q \in \Delta(\mathcal{V})} \mathrm{KL}(q \,\|\, p_\theta(\cdot \mid x_{<t})) \quad \text{subject to} \quad \langle q, \phi \rangle = b_t, \quad \mathrm{supp}(q) \subseteq \mathrm{supp}(m_t). \tag{2}$$

This is a KL projection onto the intersection of an affine slice with a face of the simplex. Its solution is an exponential tilt of the masked model distribution, $q_t \propto m_t \odot p_\theta \exp(\lambda\phi)$, with the multiplier λ fixed by the constraint; a one-dimensional monotone root find gives it to machine precision. Sec. S1 verifies the solve and its gradient oracle. Two properties matter downstream. The support restriction is satisfied *exactly*, because masked tokens receive zero mass by construction. The expectation constraint is satisfied exactly *when it is attainable*: if b_t lies outside the range of ϕ over the allowed tokens, no distribution on that support meets it, and the layer falls back to the unconstrained masked distribution and records the fallback.

Differentiating the solve. The Jacobian of q_t with respect to the incoming logits follows from the implicit function theorem applied to the scalar stationarity condition, and reduces to a rank-one correction of a diagonal, so the backward pass costs a few vector operations rather than a linear solve. Because the correction involves only inner products with ϕ, the Hessian-vector products needed for the multiplier's sensitivity can be batched; we measure a speedup from single precision of 1.33× to 2.64× over the batch sizes tested (Sec. S2).

What the mask does and does not do. On the crystal task the mask is a character-level grammar over a text serialization. It removes the syntactic failure mode completely, since every sampled string parses, but it is not a validity oracle; Sec. S3 separates the two effects.

4 The surrogate targets a concentrated distribution

Setting. The synthetic task is a length-8 sequence over a 4-symbol vocabulary, so all 65,536 sequences enumerate and every distribution below is computed rather than sampled: no divergence in this section is an estimate. A per-symbol multiplicity budget ("Regime S") defines the support constraint, satisfied by 36,120 sequences, and each symbol carries a charge; the expectation constraint asks for a mean total charge of zero.

Four arms share the mask and the architecture. **A2** trains with the mask only. **A3** trains through the projection of Eq. (2) and is read both without it (the backbone) and with it (the deployed system). **A6** is the comparator that matters: it trains by maximum likelihood on exact samples from the sequence-level tilt, which is distributional policy gradient's fixed point with unit importance weights. That training rule is chosen deliberately: it gives the comparator the best case of the method it stands for, so no gradient-estimator artefact can be blamed for the outcome.

Table 1 | Both arms satisfy the expectation constraint; only one concentrates it. The sequence-level optimum's charge variance exceeds the unconstrained data's, while the projection's is far below it. Per-sequence violation is the mass on sequences whose own total charge is nonzero. Mean ± s.d. over seeds where an arm is trained; the tilt and the data distribution are exact.

Arm	Mean charge	Violation	Variance
Data, mask-conditioned (exact)	2.5164	0.9109	8.564
A2, mask only	2.4694	0.9083	8.762
A6, trained toward the tilt	−0.0619	0.8751	10.362
Tilt, sequence-level optimum	0.0000	0.8755	10.452
A3 deployed, per-step projection	**+0.0170**	**0.6155**	**1.547**

The comparator reaches its target. A6 lands 0.0167 ± 0.0048 nats from the exact tilt with a mean charge of −0.062 ± 0.122 against a target of zero and no mass outside the mask. It is not a failed run, which is what makes the comparison in Table 1 informative rather than a win over something broken.

Result. Both A3 and A6 satisfy the expectation constraint, and A6's seed spread on mean charge is wider than the gap between them, so we claim no ordering there. The difference that lies outside seed noise is in the last two columns of Table 1. The optimum's charge variance exceeds the mask-conditioned data's, *raising* the variance of the quantity being constrained by 22.1%, while the projection lowers it by 81.9%. Against the optimum rather than against the unconstrained arm, the projection's variance is lower by a factor of 6.76, a gap exceeding three of A6's seed standard deviations. Individual samples follow: the optimum's own mass sits on constraint-violating sequences 1.42 times more often than the projection's does.

The mechanism is visible in Fig. 1 and follows from what the tilt is. It is the KL-closest distribution with the right mean, and reaching a mean of zero from a base centred at +2.469 is cheapest by moving mass symmetrically outward. Nothing in the objective rewards concentration. Bounding an expected increment at every step, by contrast, bounds the total in expectation and narrows it as a side effect, because each step's feasible set is centred on what the remaining budget can still absorb.

The surrogate's 1.025 nats of excess divergence from the optimum is therefore a trade rather than a loss. Sec. S4 reports how that overhead varies with constraint strength.

5 One checkpoint retargets across the constraint family

Because the constraint enters Eq. (1) as a target value rather than as a training condition, a trained checkpoint can be deployed at values it never trained for. We test this on the crystal task, where the constrained quantity is mass density and the family is ten density targets spanning the data distribution. Three checkpoints, each trained at one target, are deployed across all ten. Energies quoted here come from a surrogate (Sec. S5), not density functional theory.

Structural validity is exact at every one of the thirty cells, including targets no checkpoint trained for and targets beyond the reachable range. The mask is a property of the grammar, not of the

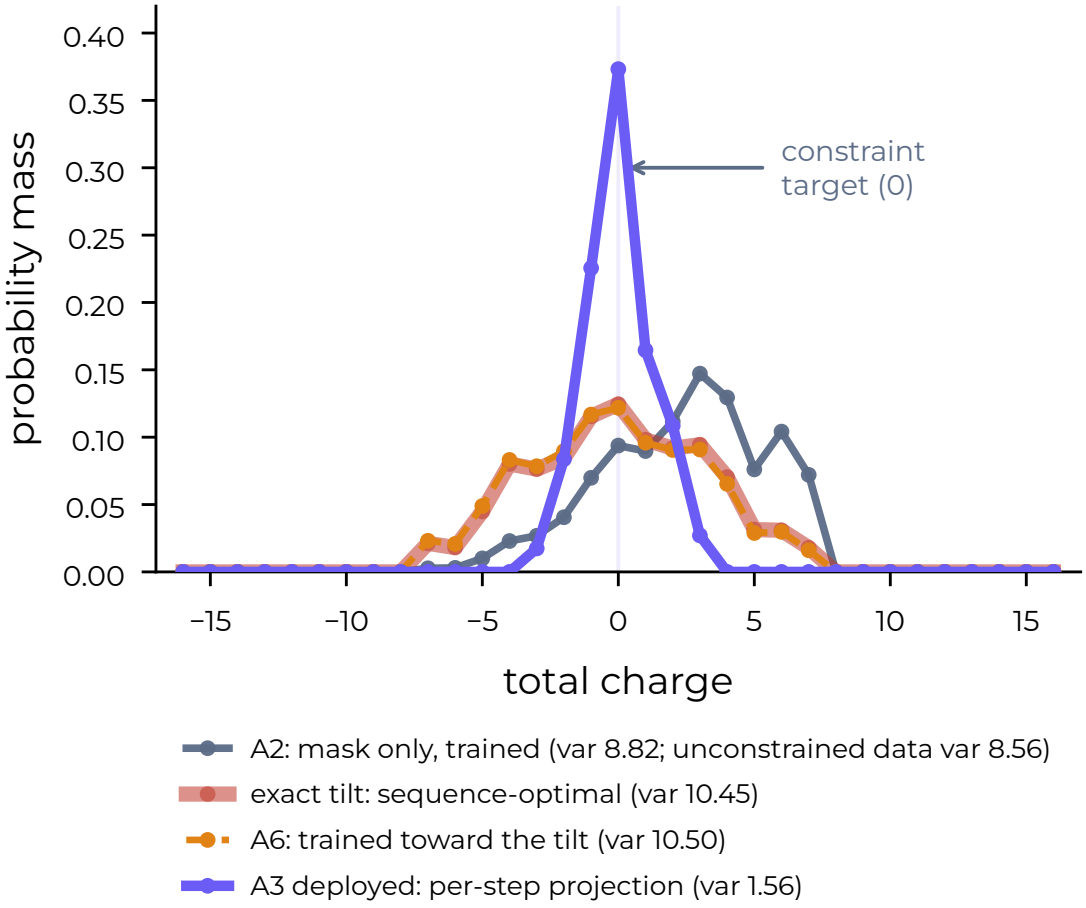
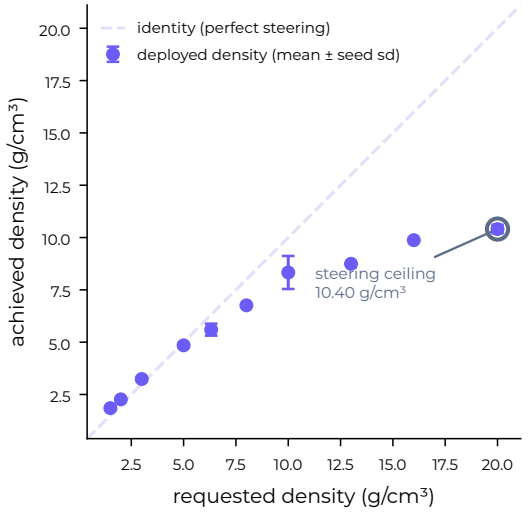


Figure 1 | The optimum spreads the constrained quantity; the projection concentrates it. Exact charge distributions over all sequences. The sequence-level optimum and the model trained toward it place mass at both extremes to buy a mean of zero, ending wider than the mask-only arm they started from. The projection instead spikes at the target. Legend entries give each arm's charge variance.

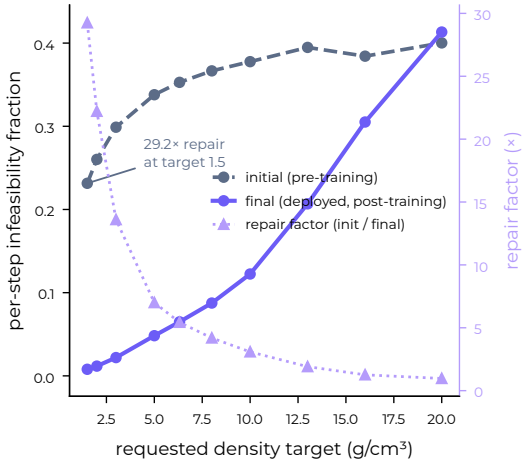
training target, so retargeting cannot break it.

On accuracy, one checkpoint deployed across the sub-ceiling family is level with the frontier of 7 separately trained models. A difference test is the wrong instrument here. A large p-value is absence of evidence, not evidence of equivalence, and at $n = 7$ paired targets the smallest attainable two-sided Wilcoxon p is $2/2^7 = 0.0156$ under any data, which Sec. S6 verifies by exhaustive enumeration. We therefore pre-registered an equivalence margin, defined as the retargeted arm's own mean within-target seed standard deviation, on the reasoning that a difference below the method's own reproducibility is not actionable and that a margin tied to the arm's own noise cannot be widened to pass without changing the arm. Two one-sided tests against that margin of 0.189 give $p = 0.0023$, and the pre-registered secondary condition ($p < 0.05$ at a 0.10 margin) also holds (Sec. S7).

The honest form of the claim is parity, not superiority: on a single seed the retargeted checkpoint



(a). Requested against achieved density, with the ceiling marked.



(b). Infeasible steps before and after training, with the repair factor.

Figure 2 | The ceiling and its mechanism. Steering follows the request until the backbone runs out of mass to redistribute; the repair the layer performs weakens as the request hardens, and where it approaches unity the ceiling binds.

appeared *better* than the frontier, but replication across seeds found it better at only 3 of 7 targets. What survives is that one training run is statistically indistinguishable from seven at one seventh of the training compute, which is a claim about cost structure rather than about accuracy. Table S1 gives the per-seed values behind both readings.

6 Where the method stops

Steering saturates at a backbone ceiling. Requested and achieved density track closely at low targets and then flatten (Fig. 2a). The plateau sits at 10.405 g/cm^3: requests beyond it are not met, and no amount of additional tilt reaches them, because the projection can only redistribute mass the backbone already assigns. The constraint is a reweighting of the model's own proposals, not a source of new modes.

Fig. 2b gives the mechanism. Before training, a large fraction of decoding steps present a budget outside the attainable range, and the layer falls back. Training repairs most of that at easy targets and progressively less at hard ones; where the repair factor approaches unity, the ceiling binds.

Two constraints compose partially, and asymmetrically. The second additive constraint is valence electron density, chosen because it is computable for every element in the split and only partly correlated with mass density ($r = 0.725$). The joint solve converges but steers weakly. Across 7 trained cells, the fraction of the requested valence shift that survives to the output never exceeds 11.3%, and the second constraint's effect is dominated by budget exhaustion: at most positions the mass budget the first constraint has already committed leaves nothing for the second to allocate.

The asymmetry is sharper than the aggregate. Asked to *lower* valence density the constraint

moves the right way in 2 of 2 cells; asked to *raise* it, in 2 of 5. A running budget is a remaining quantity: spending less of it is always available by choosing lighter or fewer atoms, while spending more than the prefix has left is not available once the budget is exhausted. Table S2 in Sec. S8 gives every cell against its own control. Charge neutrality, the constraint one would most want here, is not expressible in this class at all: it quantifies existentially over oxidation states, and collapsing to one charge per element destroys the predicate rather than approximating it (Sec. S9).

The constraint class depends on the serialization. A per-step budget requires that the prefix determine it. Mass density is a ratio, so a target density fixes a mass budget only once the cell volume is known. Over 393 structures and 1,217 element positions, a serialization that closes the cell parameters first makes the budget determinable at 100% of positions, while one that emits composition first makes it determinable at 0%. The constraint class is therefore a property of the serialization, not of the layer, and a target quantity that no prefix determines cannot be steered by this mechanism at all.

7 Baselines

Conditional training is more accurate, where it has data. Training a single model conditioned on the target value is the strongest same-cost baseline. It is more accurate than the projection wherever the training split has support: mean absolute error 0.085 against 0.196 below the data mean. It pays for that in coverage, emitting 5.9% novel formulas against 94.1%, and its accuracy degrades where the conditioning slice thins, precisely where a practitioner asking for an unusual target is operating.

Best-of-n beats the projection on accuracy, and pays in diversity. Given enough inference compute, drawing n samples from the unsteered model and keeping the closest one needs no training and wins on accuracy: it crosses the projection's error at $n = 16$ (Fig. 3). The cost is concentration of the accepted set. Unique formula fraction falls from 0.824 at $n = 1$ to 0.228 at $n = 64$, while the projection's 0.782 does not depend on how hard the target is to hit. Sequential Monte Carlo, the principled member of the same family, reaches the projection at 6 of 10 targets within $K = 64$ particles but is worse than best-of-n at equal compute by a mean factor of 4.2; its potential width is untuned, so that figure is a lower bound on what SMC could achieve rather than a refutation of it.

The honest reading is that the projection's advantage is not accuracy per target. It is that the cost is paid once at training and never again at inference, while the inference-time family pays per output forever and concentrates its outputs to do so.

8 Discussion

The result we did not expect is that the sequence-level optimum anti-concentrates the quantity it constrains. Stated after the fact it is almost obvious, since an expectation constraint says nothing about individual draws and the cheapest way to move a mean is to move the tails. It nonetheless inverts the usual reading of a per-step surrogate as a compromise. The distinction between constraining an expectation and constraining every trajectory is old (Sec. 2); what we add is a measurement of how far apart those two objects are for a concrete generative model, and the observation that the per-step route lands closer to the second.

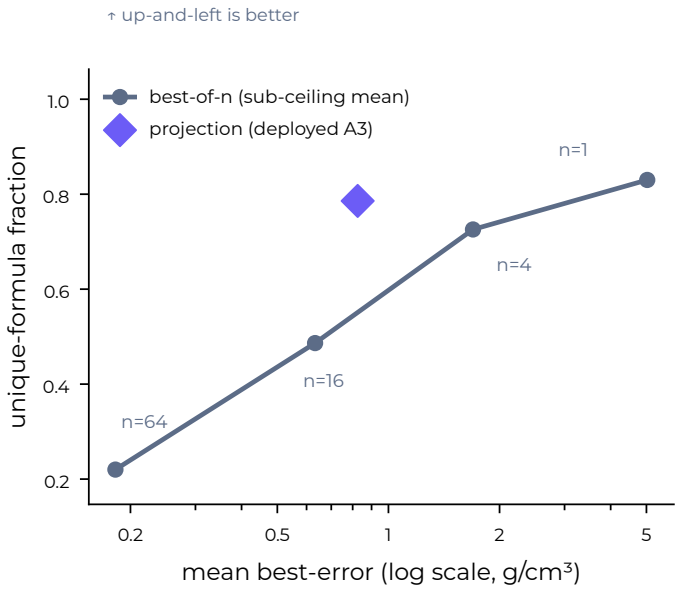


Figure 3 | **Accuracy bought by selection is diversity spent.** Mean best-error against unique-formula fraction over the sub-ceiling targets. Best-of-n walks down and to the left as n grows, trading output variety for accuracy; the projection sits off the curve at one forward pass per output.

The boundaries are as informative as the results. The steering ceiling says the projection reweights rather than creates. The composition asymmetry says a running budget is a remaining quantity, so constraints that want more of it fail differently from constraints that want less. The determinability result says the expressible constraint class is chosen when the serialization is chosen, which is a design decision usually made for other reasons. And best-of-n's accuracy win says the contribution is cost structure, not accuracy: a claim weaker than the one we set out to make and better supported than it would have been. Sec. S10 records three errors caught during this work and the mechanism that caught each, since each would otherwise have produced a plausible-looking wrong number.

9 Conclusion

Solving a small differentiable constrained projection at every decoding step, and training through it, gives an autoregressive model that satisfies an expectation constraint while concentrating the constrained quantity, something the sequence-level optimum does not do, and that retargets across a constraint family from one checkpoint. The method reweights what the backbone proposes, so it inherits a ceiling; it composes only partially when two constraints share one budget; and its expressible class is set by what the serialization makes determinable from a prefix.

> REPRODUCIBILITY
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> Every number here is read from a JSON artifact in `results/` at build time, never transcribed, and each figure and load-bearing value carries a comment naming its artifact; the test suite

> recomputes them and fails if the prose disagrees.

AUTHOR CONTRIBUTIONS

C.P. designed the layer, ran the experiments, and drafted the paper. H.K. supervised the project and revised the manuscript.

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DATA AND CODE AVAILABILITY

Code and artifacts: https://github.com/KurbanIntelligenceLab/cpt. The MP-20 subset used here is public (Sec. 2).

COMPETING INTERESTS

The authors declare no competing interests.

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APPENDIX

S1 Solver verification and the gradient oracle

The projection of Eq. (2) is solved by a one-dimensional monotone root find on the multiplier. Two things are verified rather than assumed. First, the solve converges to the constraint to machine precision wherever the target is attainable, and reports non-convergence rather than returning a

near-miss where it is not; the non-convergence flag is what the fallback path keys on. Second, the analytic Jacobian agrees with finite differences and with autograd through an unrolled solver.

For the multi-constraint form, convergence alone is insufficient: a converged solve on a near-singular system is not trustworthy. We therefore report the conditioning of the constraint matrix alongside every joint solve, and sweep vocabulary size rather than testing a single configuration, because a single grid at one vocabulary would have read as "constraints always compose".

S2 Overhead and timing

Training through the projection costs 6.0× the wall clock of training with the mask alone, on the synthetic task at matched steps. The backward pass is a rank-one correction rather than a linear solve, and batching the Hessian-vector products in single precision gives a speedup ranging from 1.33× at the smallest batch to 2.64× at the largest. At inference the layer adds one root find per step.

S3 Grammar mask and parse validity

The character-level grammar makes every sampled string parse: the syntactic parse rate is 1.00 under an independent parser. It is not a validity oracle: only 47% of those strings build into a structure, the remainder failing on semantic grounds such as overlapping sites or unphysical stoichiometries, which parse without complaint. The mask leaves real freedom: a mean of 4.14 allowed characters per step, though 70.6% of steps are fully forced by the grammar.

S4 How the overhead varies with constraint strength

The surrogate's excess divergence over the sequence-level optimum is close to fixed in absolute terms while the optimum's own cost varies over a 129-fold range as the constraint pulls harder. The relative waste therefore falls from 97.8% at the weakest steering to 32.4% at the strongest: the surrogate is least efficient exactly where the constraint barely matters, and most efficient where it binds. Sec. S1 covers the solve itself and Sec. S2 its wall-clock cost.

S5 Energy screen calibration

Formation energies are estimated with CHGNet [21] at version v0.3.0 [22] rather than density functional theory. On 1,456 held-out structures the surrogate agrees with its reference at 0.0210 eV/atom mean absolute error and Pearson $r = 0.9996$. This is a surrogate screen and is labelled as such wherever it is used; no claim in the main text depends on a DFT-quality energy.

S6 Statistical floor of the difference test

The claim that a paired Wilcoxon test at $n = 7$ cannot return $p < 0.0156$ is verified by exhaustive enumeration over all 2^7 sign patterns rather than by appeal to a table. This is why the equivalence procedure of Sec. S7, rather than the difference test, carries the retargeting claim.

Table S1 | Per-seed retargeting summary. Densities in g/cm^3. Seed 0's backbone is an excursion, which is why its lower error is not the reading to take; the published ceiling is 10.405. Structural validity is 1.0 in every cell of every seed.

Seed	Backbone mean	Max achieved	Sub-ceiling MAE
0	22.769	15.495	0.4406
1	8.381	10.439	0.7889
2	8.692	10.596	0.7197
Mean of seeds	–	–	0.6497
Frontier, 7 runs	–	–	0.6620

Table S2 | Direction of the second constraint's effect. Cells at a shared valence target across density rows. "Asked" is the direction the target sits relative to that cell's control. Lowering the quantity succeeds in both cells that request it; raising it succeeds in two of five.

Valence	Density	Achieved	Control	Deviation	Asked	Correct
0.282	3.0	0.2447	0.2424	0.0023	up	yes
0.282	5.0	0.3062	0.3141	−0.0079	down	yes
0.282	8.0	0.3251	0.3956	−0.0705	down	yes
0.484	3.0	0.2603	0.2424	0.0179	up	yes
0.484	8.0	0.3482	0.3956	−0.0474	up	no
0.686	3.0	0.2353	0.2424	−0.0071	up	no
0.686	8.0	0.3707	0.3956	−0.0248	up	no

S7 The full equivalence appendix

The margin is defined before the test, in a pre-registration file that the analysis script reads; the script fails if the file is absent or if its value disagrees with the stated rule. The rule is that the margin equals the retargeted arm's mean within-target seed standard deviation. Two one-sided tests give $p = 0.0023$ at that margin of 0.189, and the secondary pre-registered condition, still $p < 0.05$ at a 0.10 margin, also holds. A bootstrap interval is reported alongside the parametric one and the wider of the two is what the main text quotes.

The difference test is reported for completeness and interpreted as uninformative: the paired Wilcoxon over 7 targets returns $p = 1.0$, but the smallest attainable two-sided p at that sample size is $2/2^7 = 0.0156$ under any data whatsoever, so a large value carries almost no information.

S8 Two-constraint grid, per cell

Table S2 gives every cell of the two-constraint grid at a shared valence target, each against its own single-constraint control. The control is necessary: the achieved valence density rises with the density target even when no valence constraint is applied, so an absolute level would show apparent steering where there is none. Only the deviation from control is the second constraint's effect.

S9 The boundary of the expressible class

Charge neutrality cannot be written as a per-step linear expectation over this vocabulary. The screen that defines it quantifies existentially over oxidation-state assignments; most elements in the split are multivalent; and collapsing to a single charge per element destroys the predicate rather than approximating it. This is a property of the constraint, not a gap in the implementation, and it bounds the class the layer addresses. Valence electron density was adopted in its place because it is computable for every element present and only partly correlated with mass density ($r = 0.725$), so a joint solve is not one constraint written twice.

A second boundary is the serialization. Over 393 structures and 1,217 element positions, a cell-first serialization makes the density budget determinable at 100% of positions and a composition-first serialization at 0%.

S10 Defects found, and how

Three classes of error were caught during this work by mechanisms worth reporting, since each would have produced a plausible-looking wrong number.

A budget generalization initially read cell-parameter token payloads as physical values when they are bin indices, which inflated volumes while still producing budgets that looked reasonable; it was caught by an equivalence check against the original single-constraint path rather than by inspection. A charge-variance comparison was made against the unconstrained arm when the claim required the sequence-level optimum: a true statement answering a different question, caught in review, and the corrected figure is stronger than the original. A crossing count in an earlier draft was a literal in the document-generating script rather than a value computed from the data, and survived a verification pass that checked every number in the same table; it is now recomputed and asserted by a test.

The general lesson we would pass on is that verifying numbers is not the same as verifying prose. A test that recomputes a table's values will not catch the sentence describing them.